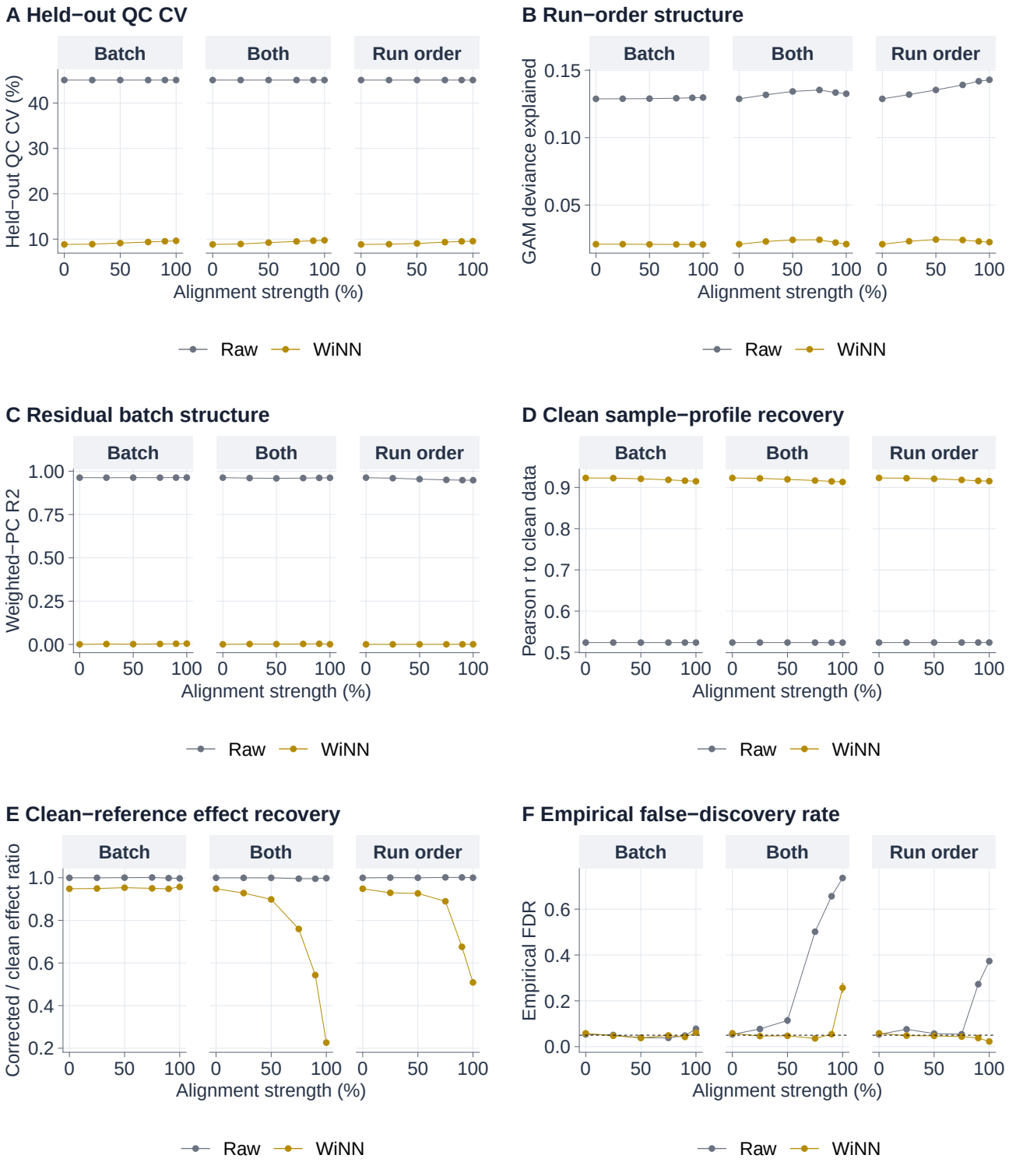

A Held−out QC CV
Batch
Both
Run order
Held−out QC CV (%)
10
20
30
40
0
50
100
Alignment strength (%)
Raw
WiNN
B Run−order structure
GAM deviance explained
0.05
0.10
0.15
C Residual batch structure
Weighted−PC R2
0.00
0.25
0.50
0.75
1.00
D Clean sample−profile recovery
Pearson r to clean data
0.5
0.6
0.7
0.8
0.9
E Clean−reference effect recovery
Corrected / clean effect ratio
0.2
0.4
0.6
0.8
1.0
F Empirical false−discovery rate
Empirical FDR
0.0
0.2
0.4
0.6